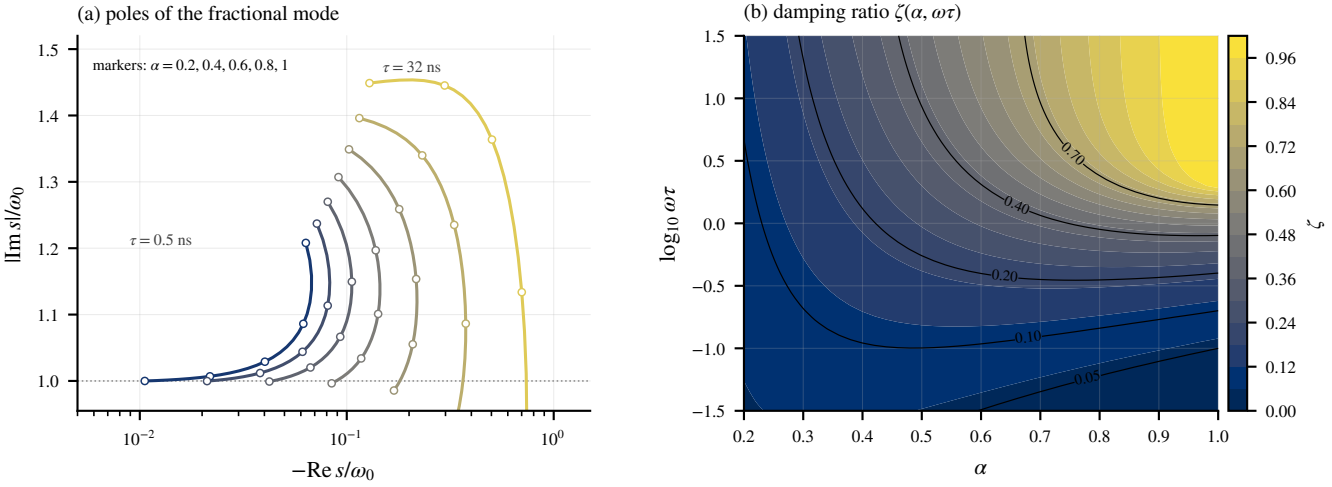
(a) poles of the fractional mode
markers: $\alpha = 0.2, 0.4, 0.6, 0.8, 1$
$\tau = 32$ ns
$\tau = 0.5$ ns
$\mathrm{Im}\, s/\omega_0$
$-\mathrm{Re}\, s/\omega_0$
(b) damping ratio $\zeta(\alpha, \omega\tau)$
$\log_{10} \omega\tau$
$\alpha$
$\zeta$
0.70
0.40
0.20
0.10
0.05